

EXERCISES



PART A – COMPETENCY FOCUSED QUESTIONS

I. Multiple Choice Questions

Tick (✓) the correct answer:

1.

int a	a=0;
float b	b=0.0f;
long p	p=0;
double k	k=0.0d;

What does the above table illustrate in Java?

- (a) Default value of data types (b) Dynamic value of data types
(c) Original value of data types (d) Unique value of data types
2. A constant which gives the exact representation of data is called
(a) Variable (b) Literal (c) Identifier (d) Character
3. A word used in a high level language which has a special meaning for the system compiler is called
(a) Class (b) Identifier (c) Keyword (d) Literal
4. A character literal is assigned to a:
(a) Char variable (b) Char type literal
(c) String variable (d) String literal
5. A character literal is enclosed in:
(a) ' ' (b) " " (c) : : (d) { }
6. A set of characters is assigned to:
(a) String variable (b) Static variable
(c) Boolean variable (d) None of these
7. The ASCII codes of uppercase letters range from:
(a) 65 – 90 (b) 60 – 85 (c) 65 – 91 (d) 97 – 122
8. Which of the following results in integer type?
(a) 11.4F/3.2D (b) 13.8F/4.6F; (c) 12/3 (d) none of these
9. Which of the following is non-primitive data?
(a) char (b) long (c) string (d) short
10. Which of the following type is an exact representation of fractional values?
(a) char (b) double (c) byte (d) String
11. Boolean data is used to test a particular condition i.e. true or false. Which of the following is a correct representation?
(a) boolean m=true; (b) boolean m='true';
(c) boolean m="true"; (d) none of these
12. Read the following text and choose the correct answer:
They are the words which are preserved by the system and carry special meaning for the system compiler. We cannot use them as names for variables or identifiers.
Which of the following characteristics of Java programming are being described?
(a) Separators (b) Literals (c) Characters (d) Keywords

II. Fill in the blanks:

1. The character sets of Java is like of English language.
2. A standard encoding system of representing characters is
3. code is decimal number to represent a character.
4. Each individual component of a Java statement is known as
5. In Java, the constants are also called
6. operator is used to store a value in the variable.
7. The comma, exclamation, question mark etc. are termed as in Java language.
8. An element of Java program that is used to identify a class, function or value is called as
9. Integer type value occupies bytes in the memory.
10. A Java expression that contains all the elements of same data type is expression.

III. Assertion and Reasoning based questions:

Assertion (A): When a variable gets initialised at run time, it is termed as 'Dynamic Initialisation'.

Reason (R): In this situation, a variable is assigned with the outcome of any arithmetical operation or function.

Based on the above discussion, choose an appropriate statement from the options given below:

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is a correct explanation of Assertion (A).
- (b) Both Assertion (A) and Reason (R) are true and Reason (R) is not a correct explanation of Assertion(A).
- (c) Assertion (A) is true and Reason (R) is false.
- (d) Assertion (A) is false and Reason (R) is true.

IV. Application based questions:

There are some non-graphical characters used as commands to direct the cursor while printing. Each individual component of a Java statement is referred to as a token. Some of the tokens are referred to as identifiers which are used to identify an element of Java program. We can also use assignment statement to store a literal, the result of an expression or the outcome of a function.

Based on the above discussion, answer the following questions:

- (a) What is not true for non-graphic characters?
 - (i) They are escape sequences.
 - (ii) They are backslash characters.
 - (iii) They control the movement of cursor while printing.
 - (iv) They are boolean literals.
- (b) Which of the following is not a token?
 - (i) Operator
 - (ii) Punctuator
 - (iii) Statement
 - (iv) Literal
- (c) Which is not the correct syntax of assignment statement?
 - (i) <expression> = <value/variable>
 - (ii) <variable> = <literal>
 - (iii) <variable> = <constant>
 - (iv) <variable> = <expression>

- (d) Which of the following is not a valid non-graphical character?
(i) "\\n" (ii) "\\t" (iii) "\\f" (iv) "\\p"

PART B – SUBJECTIVE QUESTIONS

I. Write short answers:

1. What do you mean by data type?
2. Define variable with an example. [ICSE 2006]
3. What do you mean by constant? Explain with an example.
4. State two kinds of data types. [ICSE 2006]
5. What do you understand by Token? Name different types of tokens.
6. What are the rules to assign a variable in a Java programming?
7. Explain the term 'type casting'? [ICSE 2007]
8. Perform the following:
 - (a) Assign the value of π (3.142857.....) to a variable with the requisite data type. [ICSE 2007]
 - (b) Assign the value of $\sqrt{3}$ (1.732) to a variable with the requisite data type.
9. Distinguish between:
 - (a) Integer and Floating constant
 - (b) Token and Identifier [ICSE 2008]
 - (c) Character and String constant
 - (d) Character and Boolean literal [ICSE 2009]
10. Write down the data type of the following:
 - (a) Integer
 - (b) Long Integer
 - (c) A fractional number
 - (d) A special character
11. What do you understand by Boolean data type? Explain with an example.
12. What do you understand by primitive data type? Give two examples.
13. Why is it necessary to define data types in Java programming?
14. Define the following with an example each:
 - (a) Implicit type conversion
 - (b) Explicit type conversion
15. Define 'Coercion' with reference to type conversion.
16. What do you mean by type conversion? How is implicit conversion different from explicit conversion? [ICSE 2010]
17. In what way is static declaration different from dynamic declaration?
18. What do you mean by non-primitive data type? Give examples.
19. Predict the return data type of the following:
 - (i) `int p; double q;`
`r = p+q;`
`System.out.println(r);`
 - (ii) `float m;`
`p = m/3*(Math.pow(4,3));`
`System.out.println(p);`
20. What are the resultant data types, if the following implicit conversions are performed? Show the result with flow lines.
`int i; float f; double d; char c; byte b;`
 - (a) `i + c/b;`
 - (b) `f/d + c*f;`
 - (c) `i + f - b*c;`
 - (d) `(f/i)*c + s;`
 - (e) `i + f - c + b/d;`
 - (f) `i/c + f/b;`